

History

Vineeth N Balasubramanian

Department of Computer Science and Engineering
Indian Institute of Technology, Hyderabad



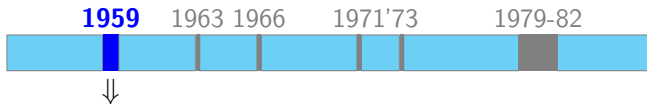
Outline

- 1 Early History: Initial Forays
- 2 Towards Algorithms and Practice: Low-level Understanding
- 3 Towards Algorithms and Practice: Next Level of Understanding
- 4 The Deep Learning Era

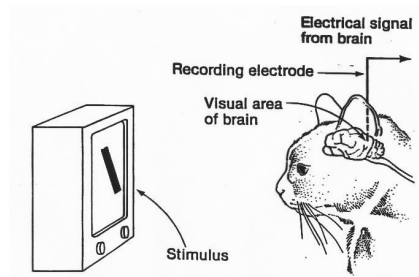
Disclaimer

- A history of the field as captured from multiple sources (including Szeliski's book and other sources credited on each page)
- A slightly biased history, as relevant to the topics we cover in this course. There is more to history in related topics (e.g. geometry-based vision, physics-based vision, image/video processing and compression, graphics, computational photography) not covered herein.
- A slight predisposition to work based on images, more than videos.

Early History¹

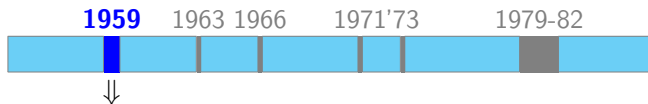


- David Hubel and Torsten Wiesel publish their work *"Receptive fields of single neurons in the cat's striate cortex"*
- Placed electrodes into primary visual cortex area of an anesthetized cat's brain
- Showed that simple and complex neurons exist, and that visual processing starts with simple structures such as oriented edges



¹Credit: [Rostyslav Demush, medium.com](https://medium.com/@rostyslavdemush)

Early History²

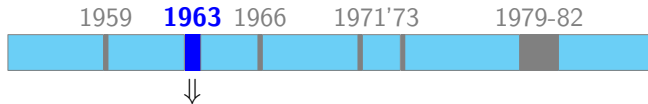


- World's first digital image: Russell Kirsch and his colleagues develop an apparatus to transform images into number grids
- Image of Russell's infant son: grainy 5cm by 5cm photo, 30,976 pixels (176×176 array)
- Now stored in Portland Art Museum



²Credit: [Rostyslav Demush, medium.com](https://medium.com/@rostyslavdemush)

Early History³



- Lawrence Roberts' PhD thesis: *"Machine Perception Of Three-Dimensional Solids"*
- Discussed extracting 3D information about solid objects from 2D photographs of line drawings
- Discussed issues such as camera transformations, perspective effects, and the rules and assumptions of depth perception

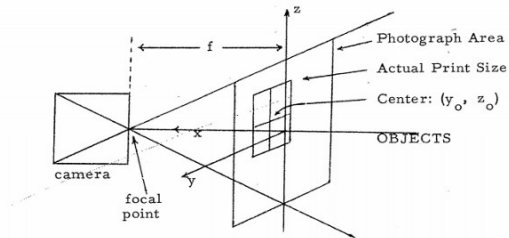
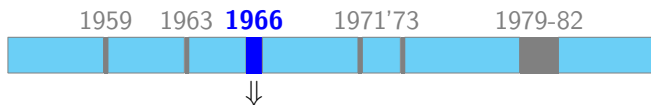


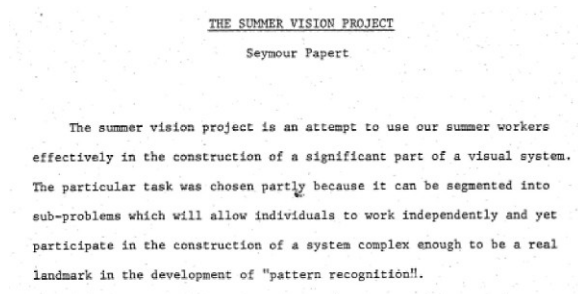
Figure 1: Camera Transformation

³Credit: [Rostyslav Demush, medium.com](https://medium.com/@rostyslav-demush)

Early History⁴

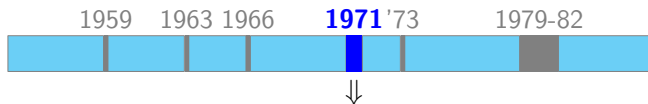


- Seymour Papert (with Gerald Sussman) from MIT launched the *Summer Vision Project*
- Aimed to develop a platform to automatically segment background/foreground and extract non-overlapping objects from real-world images

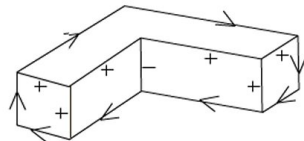


⁴Credit: [Rostyslav Demush, medium.com](https://medium.com/@rostyslav-demush)

Early History⁵

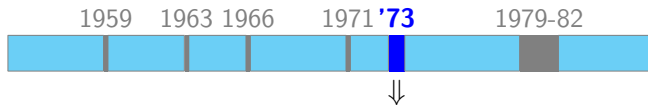


- Discern a shape in a line drawing by labeling lines as convex, concave, and occluded
 - David Huffman et al, *Impossible objects as nonsense sentences*, Machine Intelligence, 8:475-492, 1971
 - Max Clowes et al, *On seeing things*, Artificial Intelligence, 2:79-116, 1971

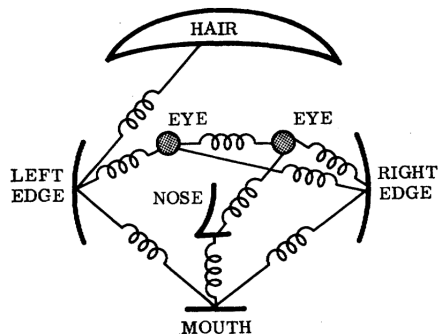


⁵Credit: [Rostyslav Demush, medium.com](https://medium.com/@rostyslav-demush/impossible-objects-as-nonsense-sentences-a7b1b1b1b1b1)

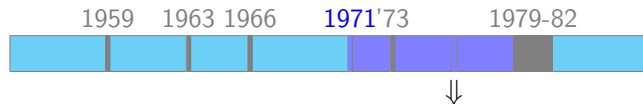
Early History



- *Pictorial Structures* model by Fischler and Elschlager
- Given a visual object's description, find the object in a photograph
- Part of the solution is specification of a descriptive scheme, and a metric on which to base the decision of "goodness" of matching or detection



Early History



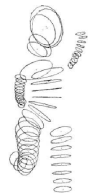
- Object recognition through shape understanding
 - Binford 1971, Generalized Cylinders
 - Marr and Nishihara 1978, Skeletons and Cylinders
- MIT's Artificial Intelligence Lab offers a "Machine Vision" course



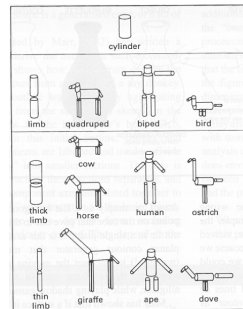
a)



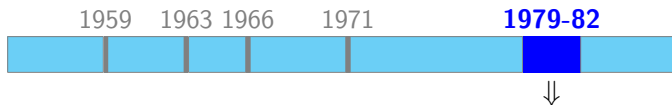
b)



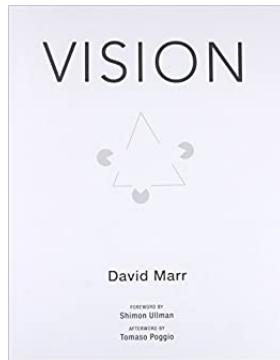
c)



Early History⁶

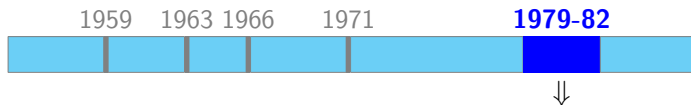


- David Marr, *Vision: A computational investigation into the human representation and processing of visual information*, 1982
- Established that vision is hierarchical
- Introduced a framework where low-level algorithms that detect edges, curves, corners, etc., are used to get high-level understanding of visual data



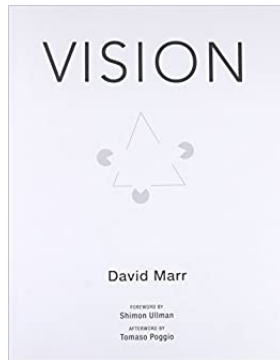
⁶Credit: [Rostyslav Demush, medium.com](https://medium.com/@rostyslavdemush)

Early History⁷



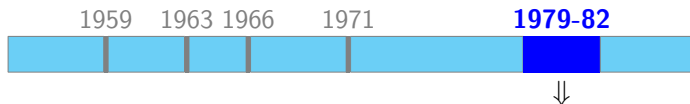
Marr's Representational Framework

- A primal sketch of an image, where edges, bars, boundaries etc., are represented
- A $2\frac{1}{2}$ -D sketch representation where surfaces, information about depth, and discontinuities on an image are pieced together
- A 3D model that is hierarchically organized in terms of surface and volumetric primitives

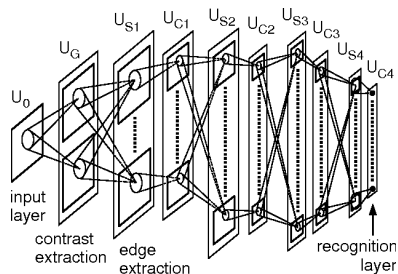


⁷Credit: [Rostyslav Demush, medium.com](https://medium.com/@rostyslav-demush)

Early History⁸



- Kunihiro Fukushima' *Neocognitron*, a self-organizing artificial network of simple and complex cells to recognize patterns, unaffected by position shifts
- The original ConvNet!
- Included convolutional layers with weight vectors (known as filters)

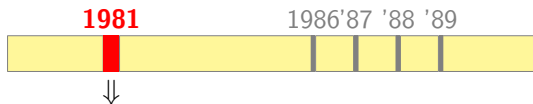


⁸Credit: [Rostyslav Demush, medium.com](https://medium.com/@rostyslav-demush)

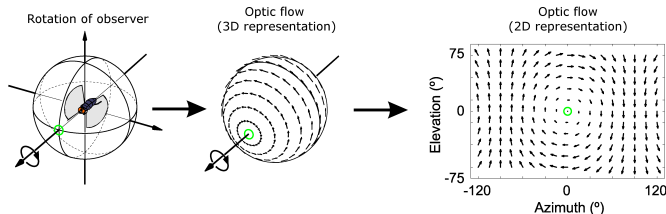
Outline

- 1 Early History: Initial Forays
- 2 Towards Algorithms and Practice: Low-level Understanding
- 3 Towards Algorithms and Practice: Next Level of Understanding
- 4 The Deep Learning Era

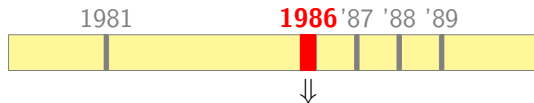
Towards Algorithms and Practice: Low-level Understanding



- **Optical Flow:** Horn and Schunck develop method to estimate the direction and speed of a moving object across two images
- Flow is formulated as a global energy functional which is minimized



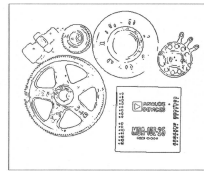
Towards Algorithms and Practice: Low-level Understanding



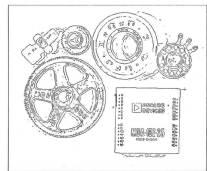
- **Canny Edge Detector:** Multi-stage edge detection operator, with a computational theory of edge detection
- Used calculus of variations to find the function that optimizes a given functional
- Well-defined method, simple to implement, became very popular for edge detection



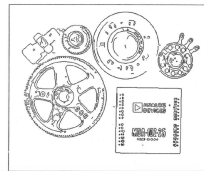
(a)



(c)

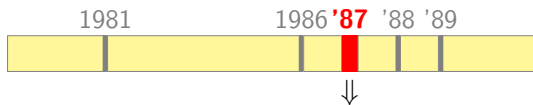


(b)

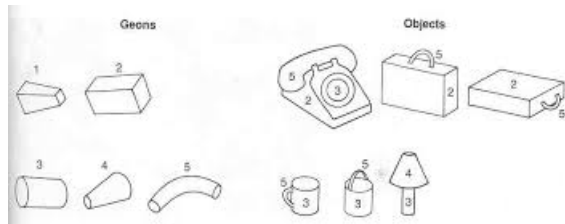


(d)

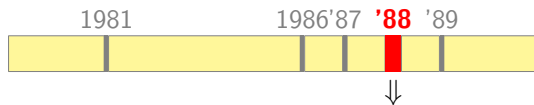
Towards Algorithms and Practice: Low-level Understanding



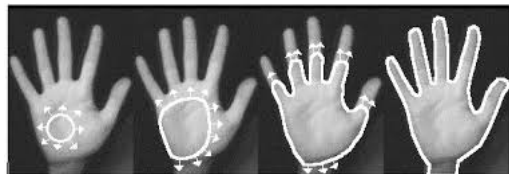
- **Recognition by Components Theory:**
Proposed by Biederman
- Bottom-up process to explain object recognition
- Object's component parts: *geons*, based on basic 3-dimensional shapes (cylinders, cones, etc.) assembled to form the object



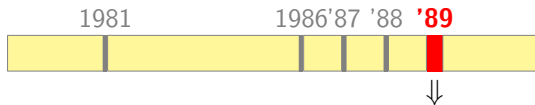
Towards Algorithms and Practice: Low-level Understanding



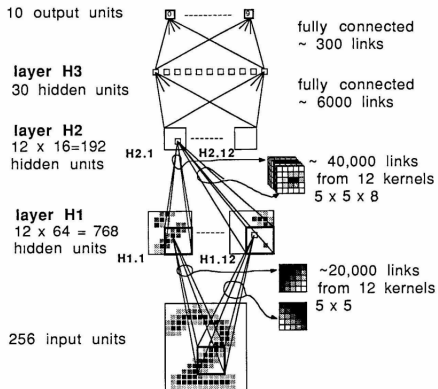
- **Snakes** or **active contour models**
delineate an object outline from a possibly noisy 2D image
- Widely used in applications like object tracking, shape recognition, segmentation, edge detection and stereo matching



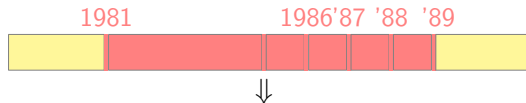
Towards Algorithms and Practice: Low-level Understanding



- **Backprop for CNNs** arrives...
- Applied to handwritten digit recognition provided by USPS



Towards Algorithms and Practice: Low-level Understanding



- Image Pyramids⁹
- Scale-space Processing¹⁰
- Wavelets¹¹
- Shape-from-X¹²
- Variational Optimization methods¹³
- Markov Random Fields¹⁴

⁹Burt and Adelson, 1983

¹⁰Witkin, 1984

¹¹Mallat, 1989

¹²Pentland, 1984; Blake et al, 1985

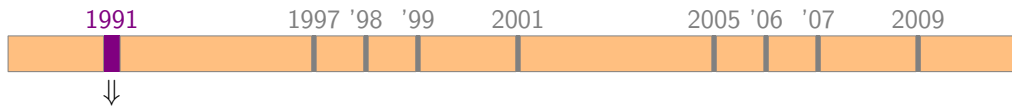
¹³Poggio et al, 1985

¹⁴Geman and Geman, 1985

Outline

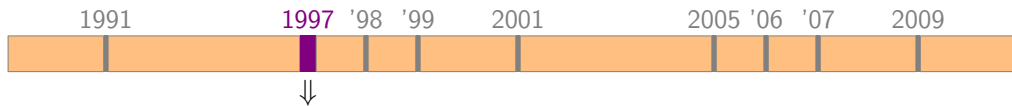
- 1 Early History: Initial Forays
- 2 Towards Algorithms and Practice: Low-level Understanding
- 3 Towards Algorithms and Practice: Next Level of Understanding
- 4 The Deep Learning Era

Towards Algorithms and Practice: Next-level Understanding



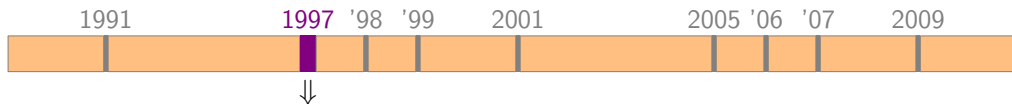
- **Eigenfaces for face recognition** (Turk & Pentland, 1991)

Towards Algorithms and Practice: Next-level Understanding



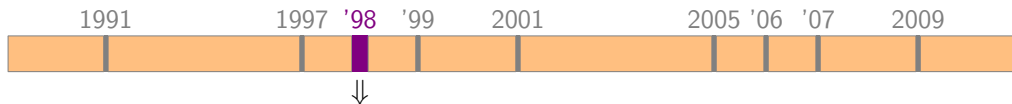
- **Eigenfaces for face recognition** (Turk & Pentland, 1991)
- **Computational theories of object recognition** (Edelman, 1997)

Towards Algorithms and Practice: Next-level Understanding



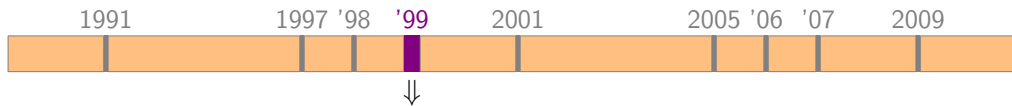
- **Eigenfaces for face recognition** (Turk & Pentland, 1991)
- **Computational theories of object recognition** (Edelman, 1997)
- **Perceptual grouping, Normalized cuts** (Shi & Malik, 1997)

Towards Algorithms and Practice: Next-level Understanding



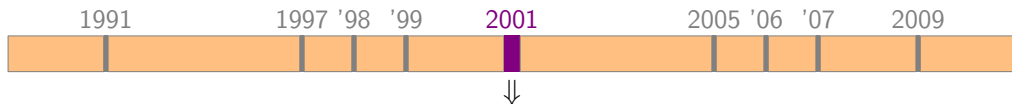
- **Eigenfaces for face recognition** (Turk & Pentland, 1991)
- **Computational theories of object recognition** (Edelman, 1997)
- **Perceptual grouping, Normalized cuts** (Shi & Malik, 1997)
- **Particle filters, Mean shift** for tracking (Liu & Chen, 1998)(Cheng, 1998)

Towards Algorithms and Practice: Next-level Understanding



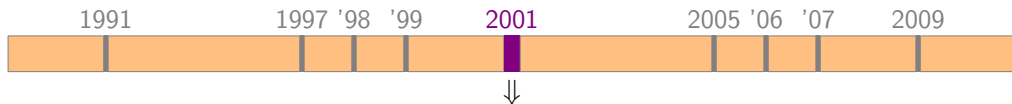
- **Eigenfaces for face recognition** (Turk & Pentland, 1991)
- **Computational theories of object recognition** (Edelman, 1997)
- **Perceptual grouping, Normalized cuts** (Shi & Malik, 1997)
- **Particle filters, Mean shift** for tracking (Liu & Chen, 1998)(Cheng, 1998)
- **SIFT** (Lowe, 1999) (Lowe, 2004)

Towards Algorithms and Practice: Next-level Understanding



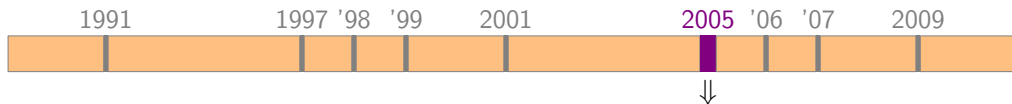
- **Eigenfaces for face recognition** (Turk & Pentland, 1991)
- **Computational theories of object recognition** (Edelman, 1997)
- **Perceptual grouping, Normalized cuts** (Shi & Malik, 1997)
- **Particle filters, Mean shift** for tracking (Liu & Chen, 1998)(Cheng, 1998)
- **SIFT** (Lowe, 1999) (Lowe, 2004)
- **Viola-Jones face detection** (Viola & Jones, 2001)

Towards Algorithms and Practice: Next-level Understanding



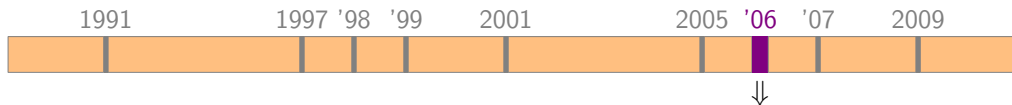
- **Eigenfaces for face recognition** (Turk & Pentland, 1991)
- **Computational theories of object recognition** (Edelman, 1997)
- **Perceptual grouping, Normalized cuts** (Shi & Malik, 1997)
- **Particle filters, Mean shift** for tracking (Liu & Chen, 1998)(Cheng, 1998)
- **SIFT** (Lowe, 1999) (Lowe, 2004)
- **Viola-Jones face detection** (Viola & Jones, 2001)
- **Conditional Random Fields** (Lafferty et al, 2001)

Towards Algorithms and Practice: Next-level Understanding



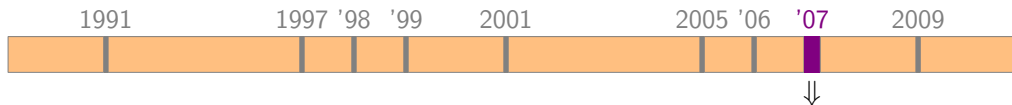
- **Eigenfaces for face recognition** (Turk & Pentland, 1991)
- **Computational theories of object recognition** (Edelman, 1997)
- **Perceptual grouping, Normalized cuts** (Shi & Malik, 1997)
- **Particle filters, Mean shift** for tracking (Liu & Chen, 1998)(Cheng, 1998)
- **SIFT** (Lowe, 1999) (Lowe, 2004)
- **Viola-Jones face detection** (Viola & Jones, 2001)
- **Conditional Random Fields** (Lafferty et al, 2001)
- **Pictorial structures** revisited (Felzenszwalb & Huttenlocher, 2005)

Towards Algorithms and Practice: Next-level Understanding



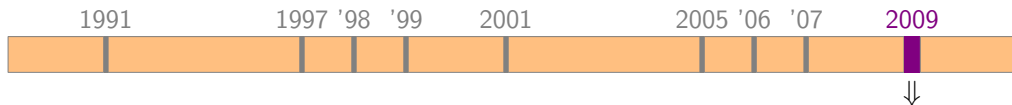
- **Eigenfaces for face recognition** (Turk & Pentland, 1991)
- **Computational theories of object recognition** (Edelman, 1997)
- **Perceptual grouping, Normalized cuts** (Shi & Malik, 1997)
- **Particle filters, Mean shift** for tracking (Liu & Chen, 1998)(Cheng, 1998)
- **SIFT** (Lowe, 1999) (Lowe, 2004)
- **Viola-Jones face detection** (Viola & Jones, 2001)
- **Conditional Random Fields** (Lafferty et al, 2001)
- **Pictorial structures** revisited (Felzenszwalb & Huttenlocher, 2005)
- **PASCAL VOC** arrives; Scene/panorama/location recognition methods grow

Towards Algorithms and Practice: Next-level Understanding



- **Eigenfaces for face recognition** (Turk & Pentland, 1991)
- **Computational theories of object recognition** (Edelman, 1997)
- **Perceptual grouping, Normalized cuts** (Shi & Malik, 1997)
- **Particle filters, Mean shift** for tracking (Liu & Chen, 1998)(Cheng, 1998)
- **SIFT** (Lowe, 1999) (Lowe, 2004)
- **Viola-Jones face detection** (Viola & Jones, 2001)
- **Conditional Random Fields** (Lafferty et al, 2001)
- **Pictorial structures** revisited (Felzenszwalb & Huttenlocher, 2005)
- **PASCAL VOC** arrives; Scene/panorama/location recognition methods grow
- **Constellation models** (Fergus, Perona & Zisserman, 2007)

Towards Algorithms and Practice: Next-level Understanding

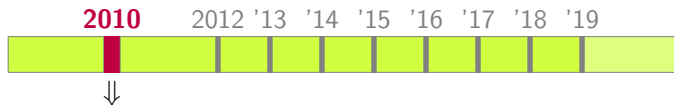


- **Eigenfaces for face recognition** (Turk & Pentland, 1991)
- **Computational theories of object recognition** (Edelman, 1997)
- **Perceptual grouping, Normalized cuts** (Shi & Malik, 1997)
- **Particle filters, Mean shift** for tracking (Liu & Chen, 1998)(Cheng, 1998)
- **SIFT** (Lowe, 1999) (Lowe, 2004)
- **Viola-Jones face detection** (Viola & Jones, 2001)
- **Conditional Random Fields** (Lafferty et al, 2001)
- **Pictorial structures** revisited (Felzenszwalb & Huttenlocher, 2005)
- **PASCAL VOC** arrives; Scene/panorama/location recognition methods grow
- **Constellation models** (Fergus, Perona & Zisserman, 2007)
- **Deformable part models** (Felzenszwalb et al, 2009)

Outline

- 1 Early History: Initial Forays
- 2 Towards Algorithms and Practice: Low-level Understanding
- 3 Towards Algorithms and Practice: Next Level of Understanding
- 4 The Deep Learning Era

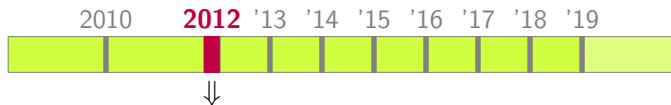
The Deep Learning Era



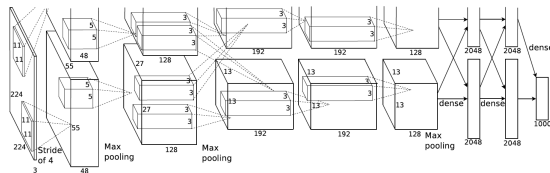
- ImageNet arrives



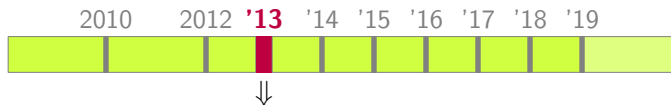
The Deep Learning Era



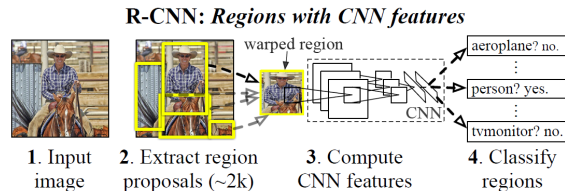
- ImageNet arrives
- AlexNet wins the ImageNet challenge



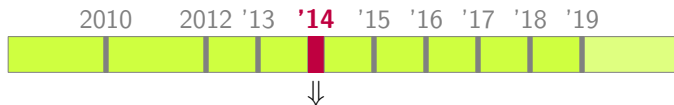
The Deep Learning Era



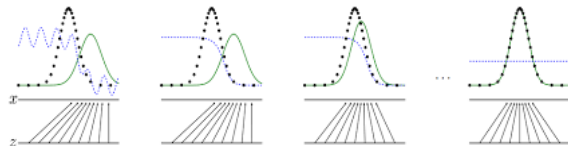
- ImageNet arrives
- AlexNet wins the ImageNet challenge
- A CNN, variant of ZFNet, wins ImageNet challenge; R-CNNs for object detection arrive; Understanding CNNs begins



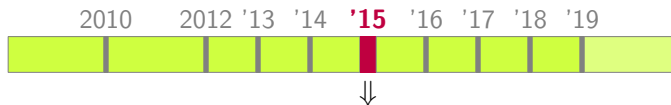
The Deep Learning Era



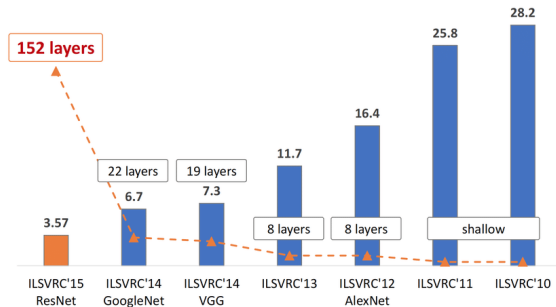
- ImageNet arrives
- AlexNet wins the ImageNet challenge
- A CNN, variant of ZFNet, wins ImageNet challenge; R-CNNs for object detection arrive; Understanding CNNs begins
- InceptionNet, VGG models arrive; Human Pose Estimation CNNs; Deep generative models: GANs, VAEs



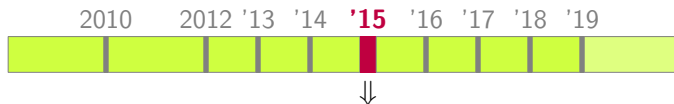
The Deep Learning Era



- ImageNet arrives
- AlexNet wins the ImageNet challenge
- A CNN, variant of ZFNet, wins ImageNet challenge; R-CNNs for object detection arrive; Understanding CNNs begins
- InceptionNet, VGG models arrive; Human Pose Estimation CNNs; Deep generative models: GANs, VAEs
- ResNet arrives; CNNs match human performance on ImageNet

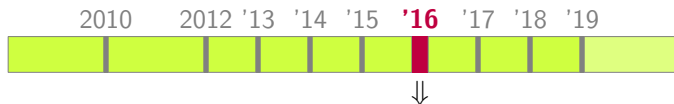


The Deep Learning Era



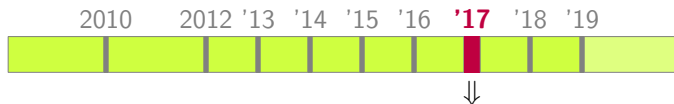
- ImageNet arrives
- AlexNet wins the ImageNet challenge
- A CNN, variant of ZFNet, wins ImageNet challenge; R-CNNs for object detection arrive; Understanding CNNs begins
- InceptionNet, VGG models arrive; Human Pose Estimation CNNs; Deep generative models: GANs, VAEs
- ResNet arrives; CNNs match human performance on ImageNet
- FCN, SegNet and U-Net for semantic segmentation; COCO dataset arrives; VQA dataset arrives

The Deep Learning Era



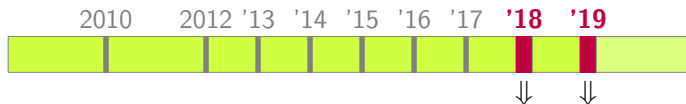
- ImageNet arrives
- AlexNet wins the ImageNet challenge
- A CNN, variant of ZFNet, wins ImageNet challenge; R-CNNs for object detection arrive; Understanding CNNs begins
- InceptionNet, VGG models arrive; Human Pose Estimation CNNs; Deep generative models: GANs, VAEs
- ResNet arrives; CNNs match human performance on ImageNet
- FCN, SegNet and U-Net for semantic segmentation; COCO dataset arrives; VQA dataset arrives
- YOLO and SSD for object detection; Cityscapes dataset arrives, Visual Genome dataset arrives

The Deep Learning Era



- ImageNet arrives
- AlexNet wins the ImageNet challenge
- A CNN, variant of ZFNet, wins ImageNet challenge; R-CNNs for object detection arrive; Understanding CNNs begins
- InceptionNet, VGG models arrive; Human Pose Estimation CNNs; Deep generative models: GANs, VAEs
- ResNet arrives; CNNs match human performance on ImageNet
- FCN, SegNet and U-Net for semantic segmentation; COCO dataset arrives; VQA dataset arrives
- YOLO and SSD for object detection; Cityscapes dataset arrives, Visual Genome dataset arrives
- Scene graph generation models

The Deep Learning Era



- ImageNet arrives
- AlexNet wins the ImageNet challenge
- A CNN, variant of ZFNet, wins ImageNet challenge; R-CNNs for object detection arrive; Understanding CNNs begins
- InceptionNet, VGG models arrive; Human Pose Estimation CNNs; Deep generative models: GANs, VAEs
- ResNet arrives; CNNs match human performance on ImageNet
- FCN, SegNet and U-Net for semantic segmentation; COCO dataset arrives; VQA dataset arrives
- YOLO and SSD for object detection; Cityscapes dataset arrives, Visual Genome dataset arrives
- Scene graph generation models
- VCR dataset, Panoptic segmentation
- ...

History of Applications¹⁵

- **1970s:** Optical Character Recognition (OCR)
- **1980s:** Machine vision, Smart cameras
- **1990s:** Machine vision in manufacturing environments, Biometrics, Medical imaging, Recording devices, Video surveillance
- **2000s:** More biometrics, Better medical imaging, Object/Face detection, Autonomous navigation, Google Goggles, Vision on social media
- **2010s:** Everywhere around us

¹⁵See <https://www.phase1vision.com/resources/timeline> for a longer historical timeline

Homework

Readings

- Chapter 1, Szeliski, *Computer Vision: Algorithms and Applications* (Sections 1.1 and 1.2, in particular)
- Other links provided on respective slides, especially
 - [Rostyslav Demush, A Brief History of Computer Vision \(and Convolutional Neural Networks\)](#)

Relevant References I



David Huffman. “Impossible objects as nonsense sentences | AMiner”. In: *Machine Intelligence* 8 (1971), pp. 475–492.



M.A. Fischler and R.A. Elschlager. “The Representation and Matching of Pictorial Structures”. In: *IEEE Transactions on Computers* C-22.1 (Jan. 1973), pp. 67–92.



Agin and Binford. “Computer Description of Curved Objects”. In: *IEEE Transactions on Computers* C-25.4 (Apr. 1976), pp. 439–449.



D. Marr, H. K. Nishihara, and Sydney Brenner. “Representation and recognition of the spatial organization of three-dimensional shapes”. In: *Proceedings of the Royal Society of London. Series B. Biological Sciences* 200.1140 (Feb. 1978), pp. 269–294.



Kunihiko Fukushima. “Neocognitron: A self-organizing neural network model for a mechanism of pattern recognition unaffected by shift in position”. In: *Biological Cybernetics* 36.4 (Apr. 1980), pp. 193–202.



Berthold K. P. Horn and Brian G. Schunck. “Determining optical flow”. In: *Artificial Intelligence* 17.1 (Aug. 1981), pp. 185–203.

Relevant References II



P. Burt and E. Adelson. "The Laplacian Pyramid as a Compact Image Code". In: *IEEE Transactions on Communications* 31.4 (Apr. 1983), pp. 532–540.



Stuart Geman and Donald Geman. "Stochastic Relaxation, Gibbs Distributions, and the Bayesian Restoration of Images". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* PAMI-6.6 (Nov. 1984), pp. 721–741.



A. Witkin. "Scale-space filtering: A new approach to multi-scale description". In: *ICASSP '84. IEEE International Conference on Acoustics, Speech, and Signal Processing*. Vol. 9. Mar. 1984, pp. 150–153.



Tomaso Poggio, Vincent Torre, and Christof Koch. "Computational vision and regularization theory". In: *Nature* 317.6035 (Sept. 1985), pp. 314–319.



John Canny. "A Computational Approach to Edge Detection". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* PAMI-8.6 (Nov. 1986), pp. 679–698.



Irving Biederman. "Recognition-by-components: a theory of human image understanding". In: *Psychological Review* 94.2 (Apr. 1987), pp. 115–147.

Relevant References III



L. Sirovich and M. Kirby. “Low-dimensional procedure for the characterization of human faces”. In: *JOSA A* 4.3 (Mar. 1987), pp. 519–524.



Michael Kass, Andrew Witkin, and Demetri Terzopoulos. “Snakes: Active contour models”. In: *International Journal of Computer Vision* 1.4 (Jan. 1988), pp. 321–331.



Y. LeCun et al. “Backpropagation Applied to Handwritten Zip Code Recognition”. In: *Neural Computation* 1.4 (Dec. 1989), pp. 541–551.



S.G. Mallat. “A theory for multiresolution signal decomposition: the wavelet representation”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 11.7 (July 1989), pp. 674–693.



M.A. Turk and A.P. Pentland. “Face recognition using eigenfaces”. In: *1991 IEEE Computer Society Conference on Computer Vision and Pattern Recognition Proceedings*. June 1991, pp. 586–591.



Yizong Cheng. “Mean shift, mode seeking, and clustering”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 17.8 (Aug. 1995), pp. 790–799.

Relevant References IV



Pierre Del Moral. “Nonlinear filtering: Interacting particle resolution”. In: *Comptes Rendus de l'Académie des Sciences - Series I - Mathematics* 325.6 (Sept. 1997), pp. 653–658.



Jianbo Shi and Jitendra Malik. “Normalized Cuts and Image Segmentation”. In: *Proceedings of the 1997 Conference on Computer Vision and Pattern Recognition (CVPR '97)*. CVPR '97. USA: IEEE Computer Society, June 1997, p. 731.



Jun S. Liu and Rong Chen. “Sequential Monte Carlo Methods for Dynamic Systems”. In: *Journal of the American Statistical Association* 93.443 (Sept. 1998), pp. 1032–1044.



D.G. Lowe. “Object recognition from local scale-invariant features”. In: *Proceedings of the Seventh IEEE International Conference on Computer Vision*. Vol. 2. Sept. 1999, 1150–1157 vol.2.



Jianbo Shi and J. Malik. “Normalized cuts and image segmentation”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 22.8 (Aug. 2000), pp. 888–905.

Relevant References V



John D. Lafferty, Andrew McCallum, and Fernando C. N. Pereira. “Conditional Random Fields: Probabilistic Models for Segmenting and Labeling Sequence Data”. In: *Proceedings of the Eighteenth International Conference on Machine Learning*. ICML '01. San Francisco, CA, USA: Morgan Kaufmann Publishers Inc., June 2001, pp. 282–289.



Kunihiko Fukushima. “Neocognitron for handwritten digit recognition”. In: *Neurocomputing* 51 (Apr. 2003), pp. 161–180.



David G. Lowe. “Distinctive Image Features from Scale-Invariant Keypoints”. In: *International Journal of Computer Vision* 60.2 (Nov. 2004), pp. 91–110.



Paul Viola and Michael J. Jones. *Robust Real-Time Face Detection*. May 2004. URL: <https://doi.org/10.1023/B:VISI.0000013087.49260.fb> (visited on 04/08/2020).



Pedro F. Felzenszwalb and Daniel P. Huttenlocher. “Pictorial Structures for Object Recognition”. In: *International Journal of Computer Vision* 61.1 (Jan. 2005), pp. 55–79.

Relevant References VI



R. Fergus, P. Perona, and A. Zisserman. “Weakly Supervised Scale-Invariant Learning of Models for Visual Recognition”. In: *International Journal of Computer Vision* 71.3 (Mar. 2007), pp. 273–303.



Pedro F. Felzenszwalb et al. “Object Detection with Discriminatively Trained Part-Based Models”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 32.9 (Sept. 2010), pp. 1627–1645.



Richard Szeliski. *Computer Vision: Algorithms and Applications*. Texts in Computer Science. London: Springer-Verlag, 2011.



Alex Krizhevsky, Ilya Sutskever, and Geoffrey E. Hinton. “ImageNet classification with deep convolutional neural networks”. In: *Proceedings of the 25th International Conference on Neural Information Processing Systems - Volume 1*. NIPS'12. Lake Tahoe, Nevada: Curran Associates Inc., Dec. 2012, pp. 1097–1105.



Weijun Wang and Ramakant Nevatia. “Robust Object Tracking Using Constellation Model with Superpixel”. In: *Computer Vision – ACCV 2012*. Ed. by Kyoung Mu Lee et al. Lecture Notes in Computer Science. Berlin, Heidelberg: Springer, 2013, pp. 191–204.

Relevant References VII



Ross Girshick et al. “Rich Feature Hierarchies for Accurate Object Detection and Semantic Segmentation”. In: *2014 IEEE Conference on Computer Vision and Pattern Recognition*. June 2014, pp. 580–587.



Ian J. Goodfellow et al. “Generative adversarial nets”. In: *Proceedings of the 27th International Conference on Neural Information Processing Systems - Volume 2*. NIPS’14. Montreal, Canada: MIT Press, Dec. 2014, pp. 2672–2680.



Diederik P. Kingma. “Auto-Encoding Variational Bayes.”. In: (2014).



Alexander Toshev and Christian Szegedy. “DeepPose: Human Pose Estimation via Deep Neural Networks”. In: *2014 IEEE Conference on Computer Vision and Pattern Recognition*. June 2014, pp. 1653–1660.



Karen Simonyan and Andrew Zisserman. “Very Deep Convolutional Networks for Large-Scale Image Recognition”. In: *arXiv:1409.1556 [cs]* (Apr. 2015).

Relevant References VIII



Christian Szegedy et al. “Going deeper with convolutions”. In: *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. June 2015, pp. 1–9.



Kaiming He et al. “Deep Residual Learning for Image Recognition”. In: *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. June 2016, pp. 770–778.



Vidolab. *Computer Vision History: Milestones and Breakthroughs*. Apr. 2020. URL: <https://computer-vision-ai.com/blog/computer-vision-history/> (visited on 04/06/2020).



Rostyslav Demush. *A Brief History of Computer Vision (and Convolutional Neural Networks) | Hacker Noon*. URL: <https://hackernoon.com/a-brief-history-of-computer-vision-and-convolutional-neural-networks-8fe8aacc79f3> (visited on 04/06/2020).



History of Machine Vision. URL: <https://www.phase1vision.com/resources/timeline> (visited on 04/12/2020).



ImageNet. URL: <http://www.image-net.org/> (visited on 04/12/2020).

Relevant References IX



The MIT Press. *Robot Vision* | *The MIT Press*. URL:
<https://mitpress.mit.edu/books/robot-vision> (visited on 04/07/2020).



pubmeddev and Edelman S. *Computational theories of object recognition*. - *PubMed - NCBI*. URL:
<https://www.ncbi.nlm.nih.gov/pubmed/21223932> (visited on 04/08/2020).



The PASCAL Visual Object Classes Challenge 2006. URL:
<http://host.robots.ox.ac.uk/pascal/VOC/voc2006/index.html> (visited on 04/08/2020).



Zbigniew Zdziarski. *The Beginnings of Computer Vision*. URL:
<https://zbigatron.com/the-early-history-of-computer-vision/> (visited on 04/06/2020).